

lean Sila

perfectly balanced



perfectly balanced



SCALE L1
SCALE L2
IMPROVE UX

LEAN CL
LEAN DL
LEAN EL

max security
max simplicity
max longevity

part 1—tracks

part 2—principles

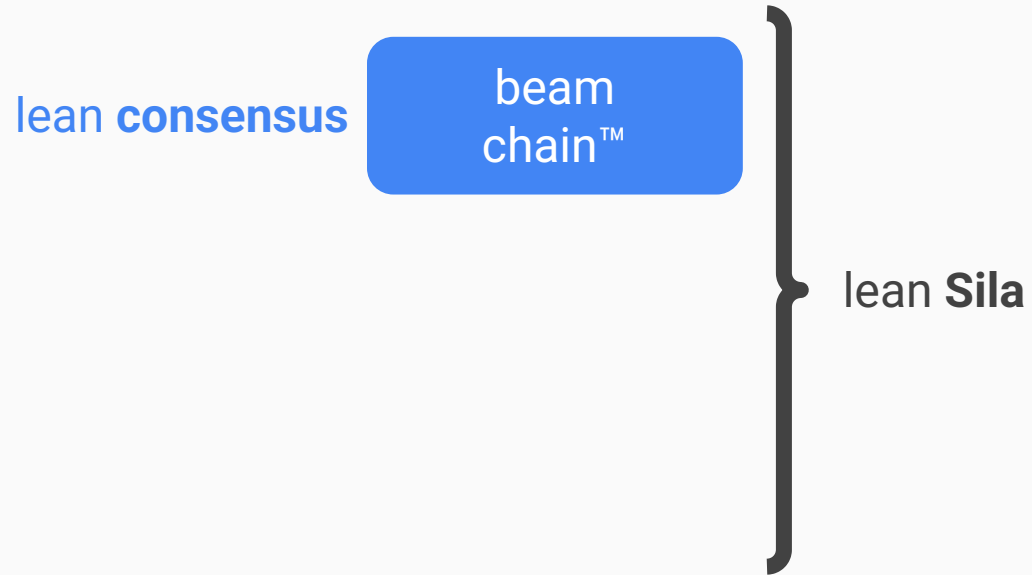
part 3—synergies

part 1—tracks

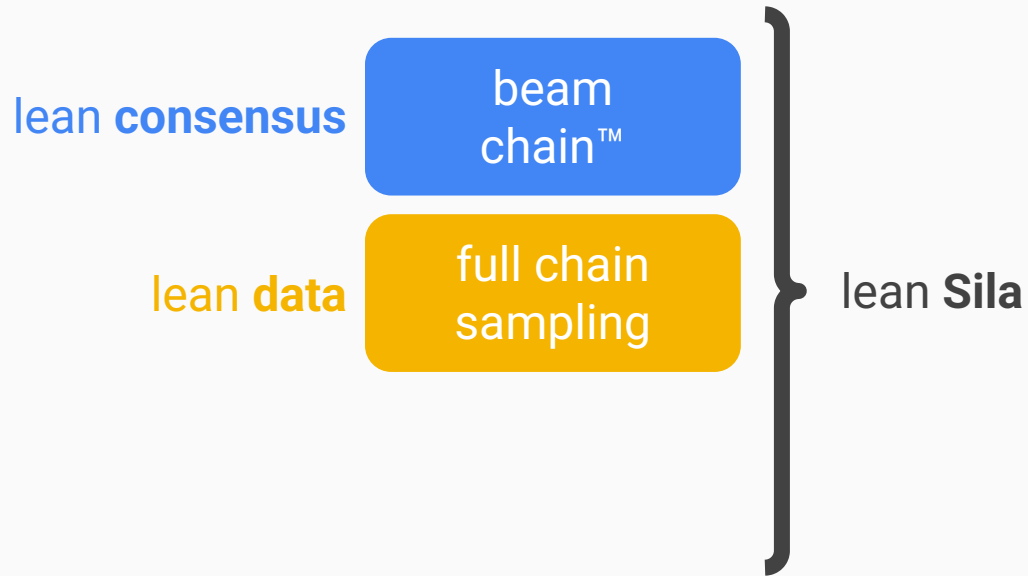
part 2—principles

part 3—synergies

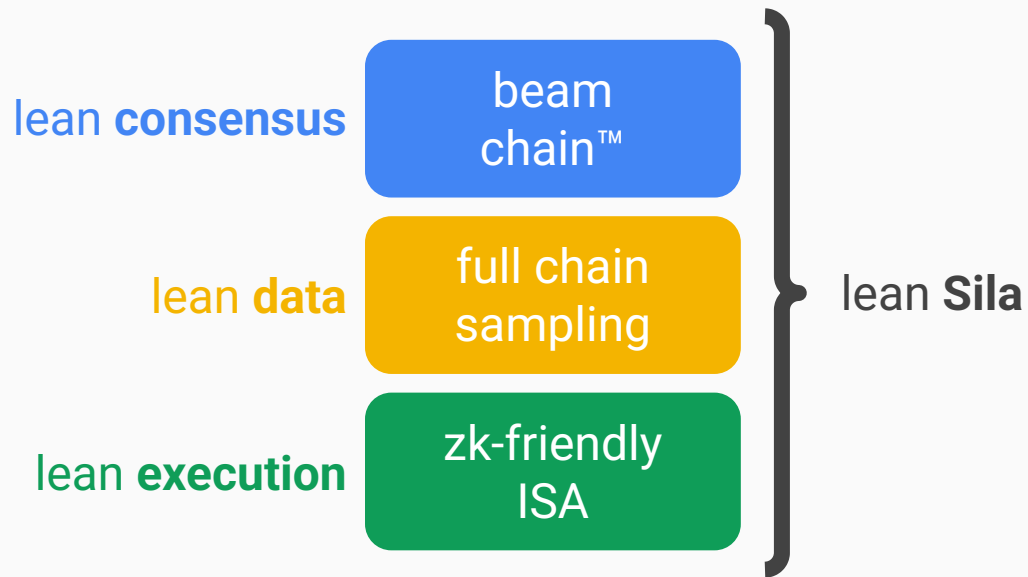
three tracks



three tracks



three tracks



headliners

PQ

snarkification

3SF

APS

PQ

snarkification

3SF

APS



hash-based
DAS

full chain
sampling

headliners

PQ

snarkification

3SF

APS



BLOBS

CALLDATA

hash-based
DAS

full chain
sampling



EVM

**LEAN
zkEVM**

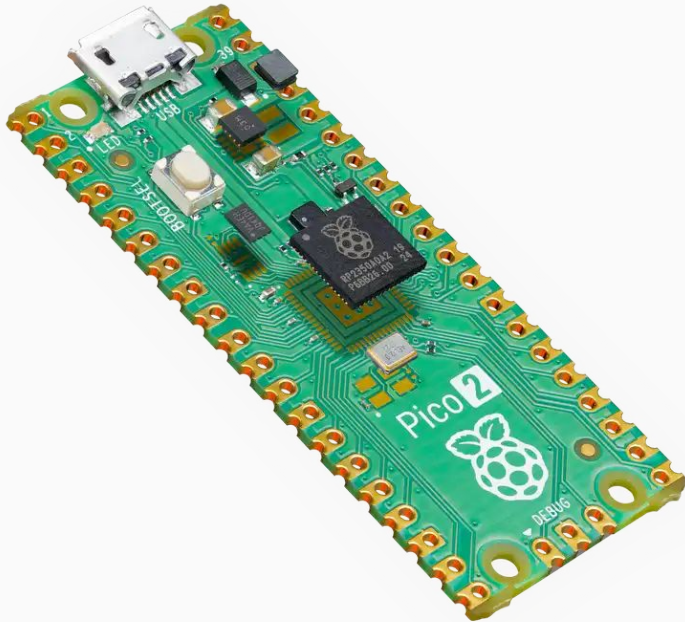
simple
zkVM

enshrined
zkVM

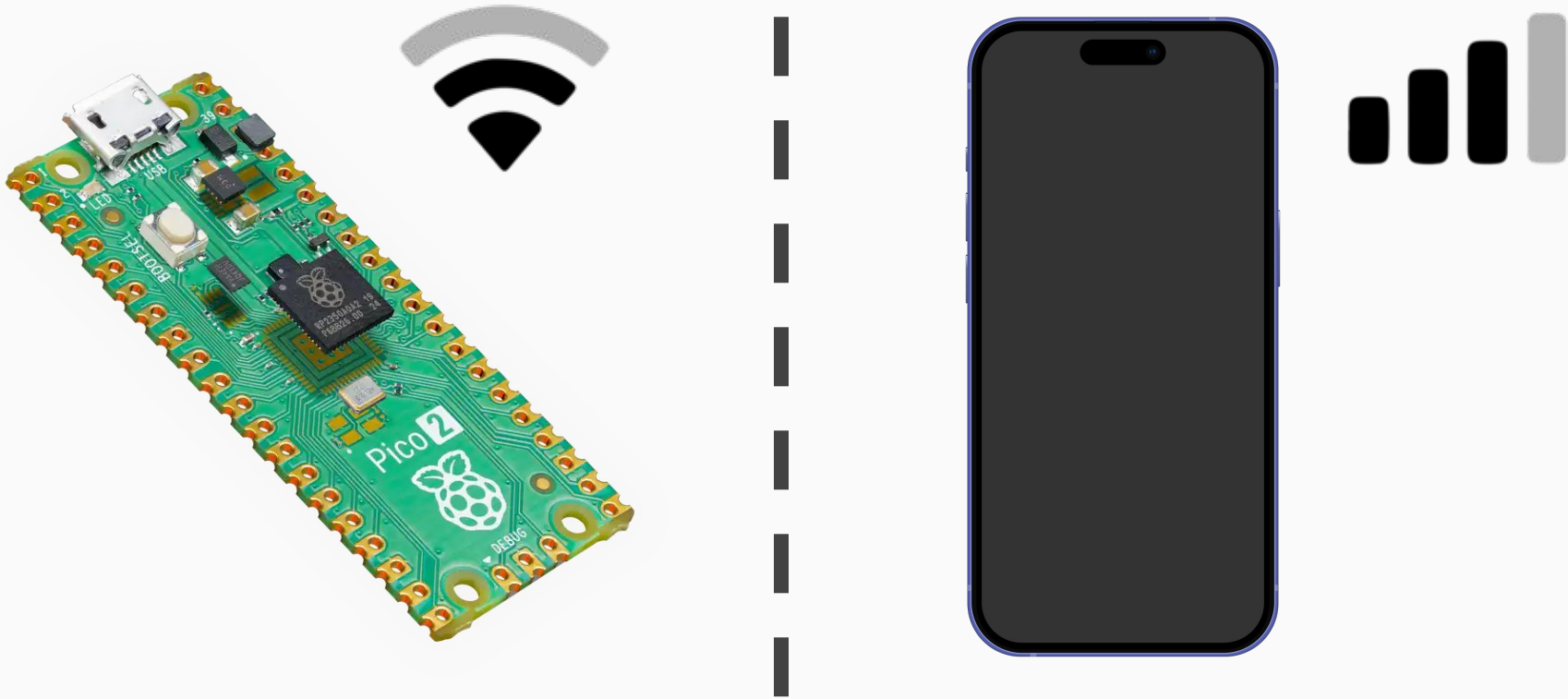
part 1—tracks

part 2—principles

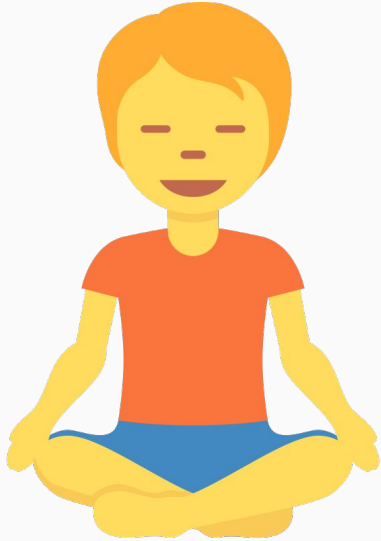
part 3—synergies



validators, wallets, includers

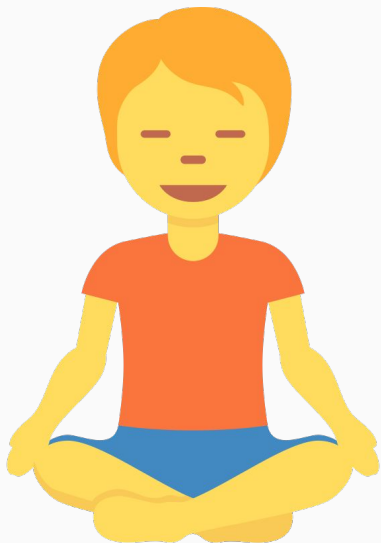


validators, wallets, includers



attester-proposer
separation

-
-
-
-



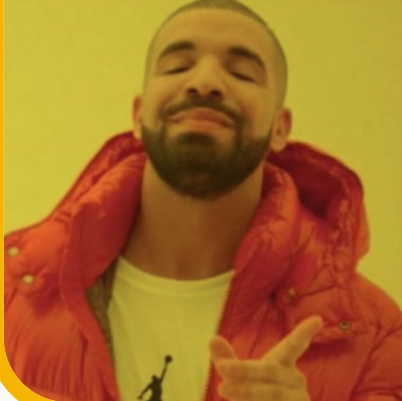
attester-proposer
separation

- no relays
- no timing games
- no MEV spikes
- no preconf

lean assumptions

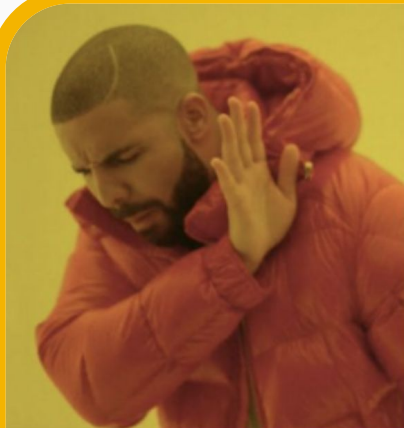


**BLS
KZG**

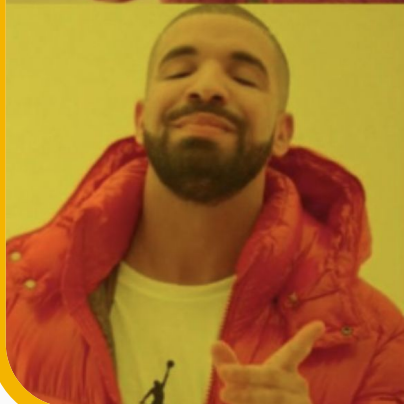


PQ

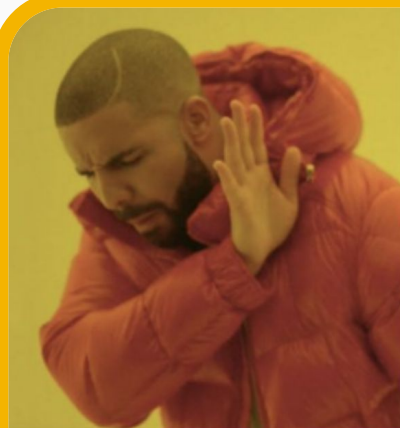
lean assumptions



**BLS
KZG**



PQ



**LATTICES
ISOGENIES**

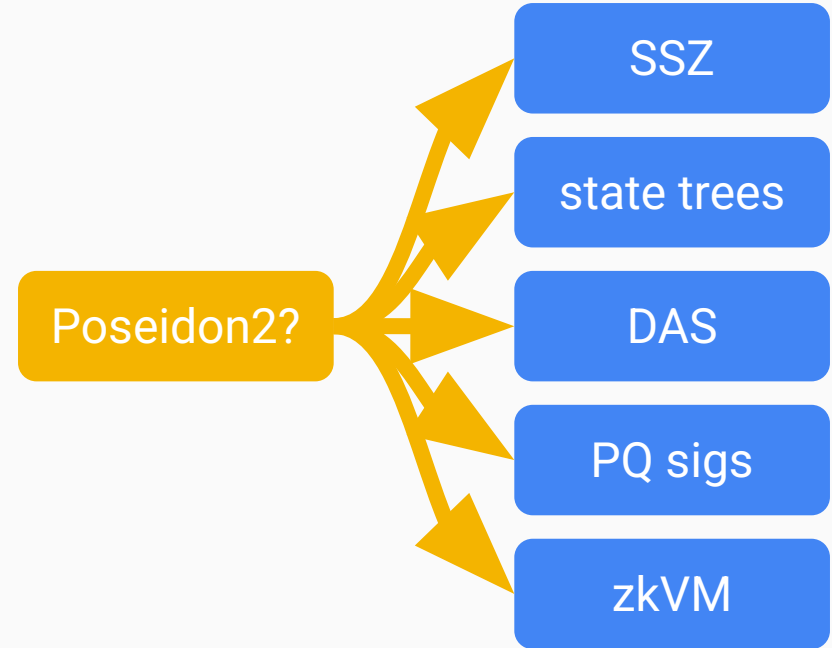


HASHES



Poseidon2?





lean **finality**

112 LoC

[sila/research/3sf-mini](#)

lean **snarks**

445 LoC

[TomWambsgans/simple-snark-spec](#)

lean **ISA**

51 LoC

[ChatGPT: "minimal Python RISC-V"](#)

lean **signatures**

57 LoC

[ChatGPT: "minimal WOTS XMSS"](#)

lean **finality**

112 LoC

[sila/research/3sf-mini](#)

lean **snarks**

445 LoC

[TomWambsgans/simple-snark-spec](#)

encapsulated
complexity

lean **ISA**

51 LoC

[ChatGPT: "minimal Python RISC-V"](#)

aesthetics

lean **signatures**

57 LoC

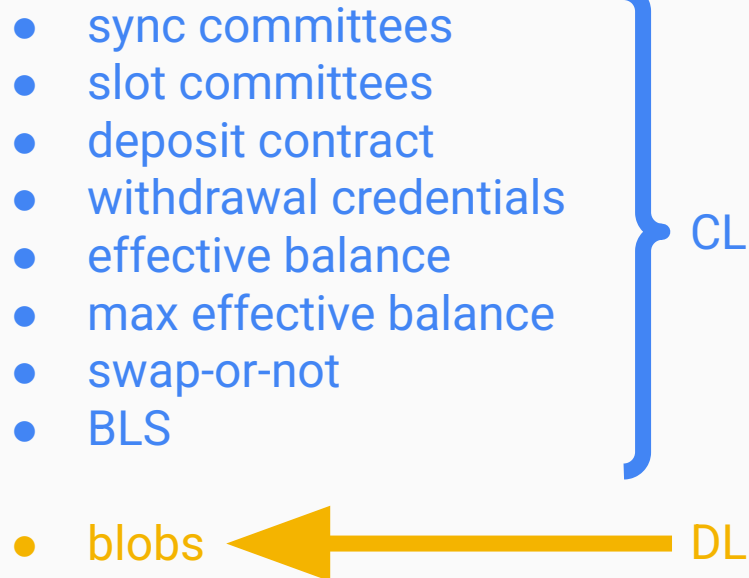
[ChatGPT: "minimal WOTS XMSS"](#)

lean tech debt

- sync committees
- slot committees
- deposit contract
- withdrawal credentials
- effective balance
- max effective balance
- swap-or-not
- BLS



lean tech debt

- sync committees
 - slot committees
 - deposit contract
 - withdrawal credentials
 - effective balance
 - max effective balance
 - swap-or-not
 - BLS
- CL
- blobs
- DL
- 
- The diagram shows a list of nine items. The first eight items are grouped by a blue bracket on the right, with the label 'CL' next to it. A yellow arrow points from the 'blobs' item to the 'DL' label.



lean tech debt

- sync committees
- slot committees
- deposit contract
- withdrawal credentials
- effective balance
- max effective balance
- swap-or-not
- BLS

CL

• blobs ← DL

• EVM ← EL



improve status quo
vs
reach endgame

improve status quo
vs
reach endgame

optimality

3SF

Hypercube

minslack

improve status quo
vs
reach endgame

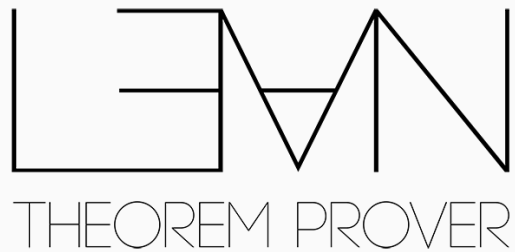
optimality

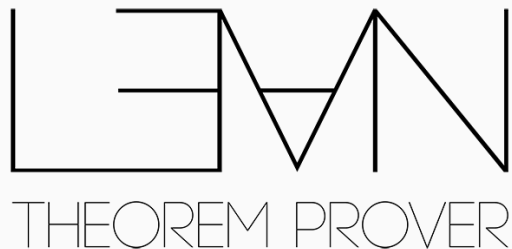
3SF

Hypercube

minslack







SILA

zkEVM Formal Verification

verified-zkevm.org

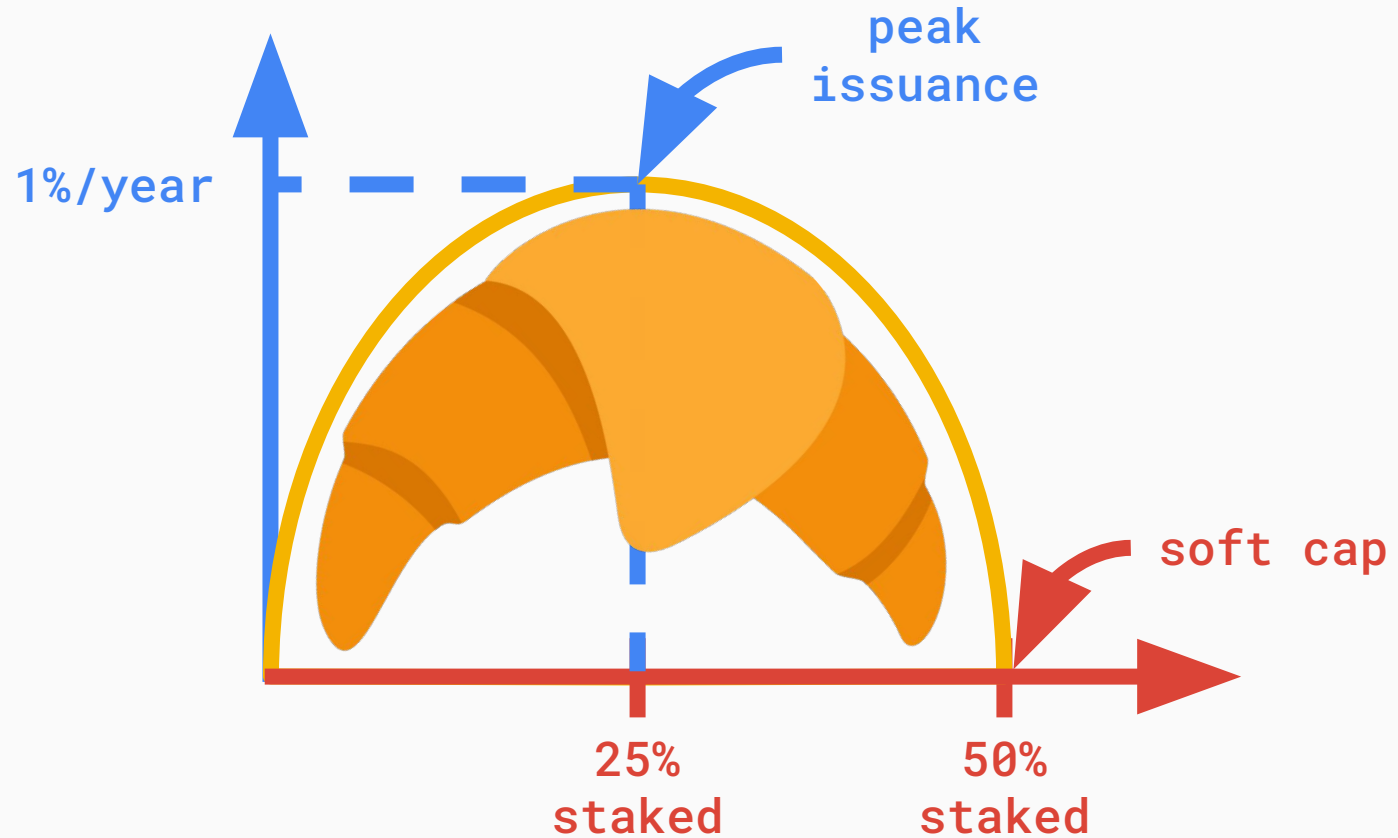
3SF Automated Model-Checking

freespek/ssf-mc

lean issuance



lean issuance

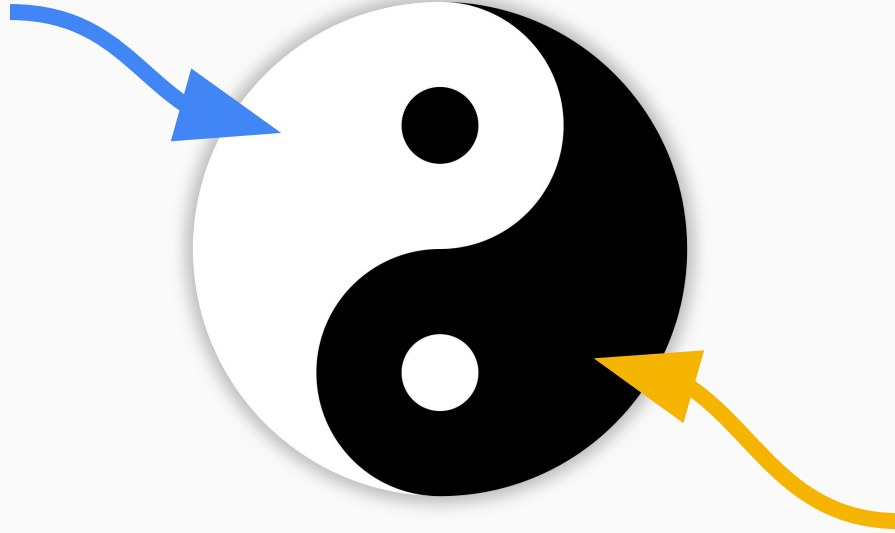


part 1—tracks

part 2—principles

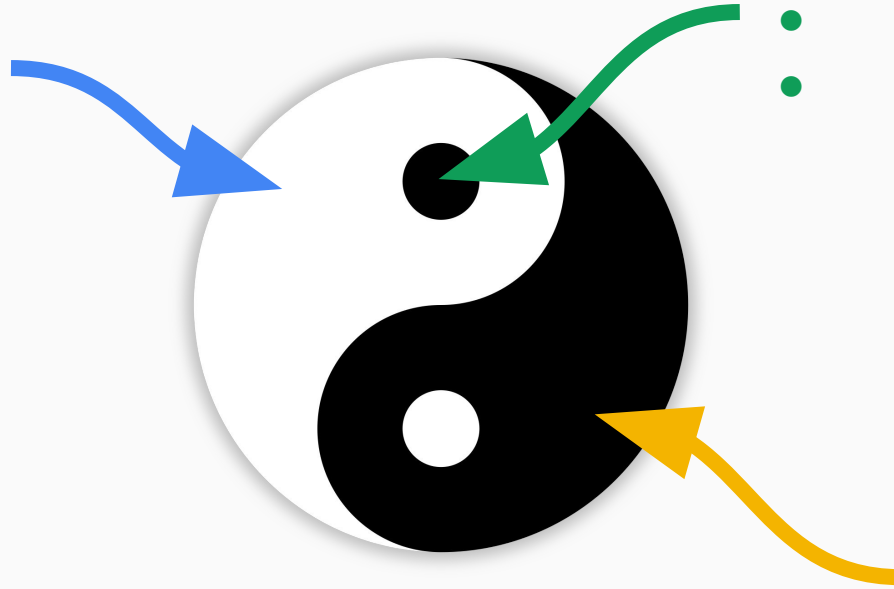
part 3—synergies

- gigagas L1
- teragas L2
- amazing UX



- \$1Q security
- optimality
- ossification

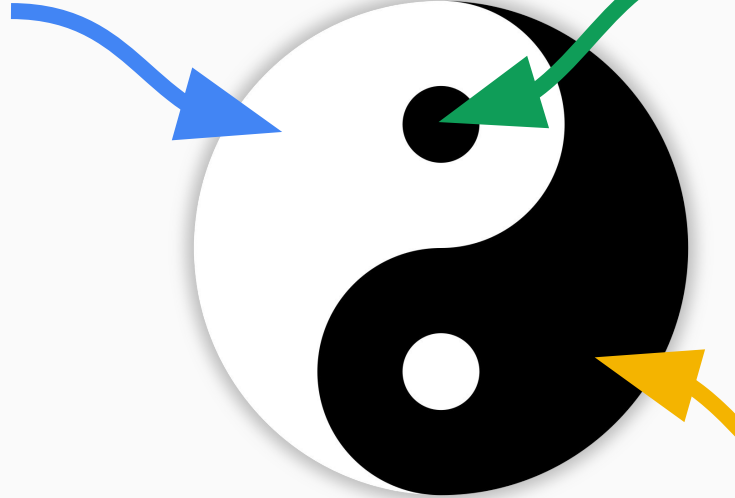
- gigagas L1
- teragas L2
- amazing UX



- FOCIL
-
-

- \$1Q security
- optimality
- ossification

- gigagas L1
- teragas L2
- amazing UX

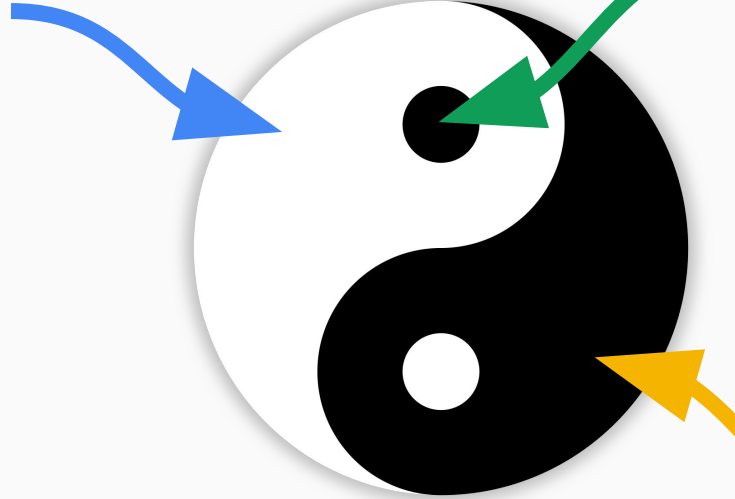


- FOCIL
- zkEVMs
-

- \$1Q security
- optimality
- ossification

lean synergies

- gigagas L1
- teragas L2
- amazing UX



- FOCIL
- zkEVMs
- FCS

- \$1Q security
- optimality
- ossification

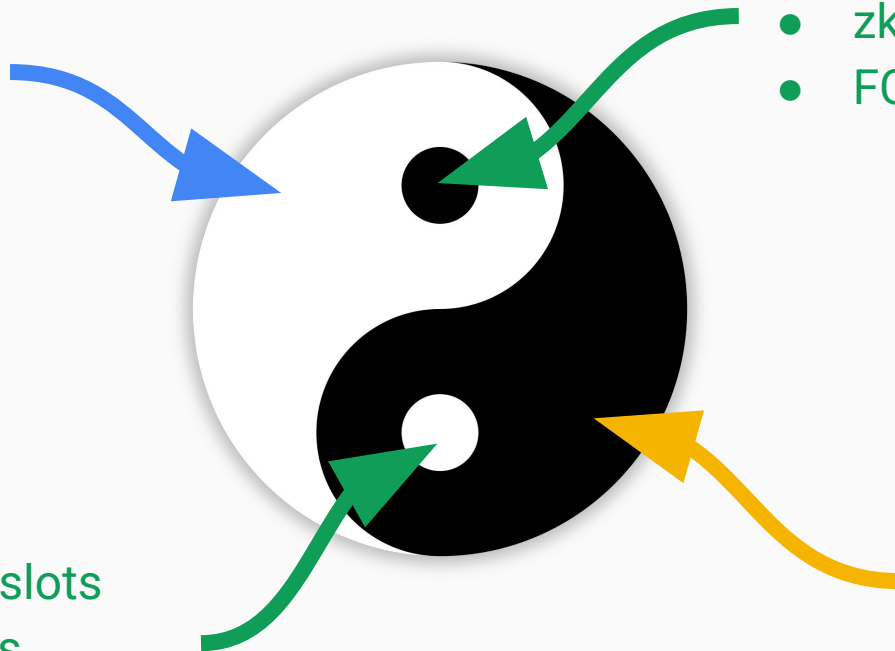
lean synergies

- gigagas L1
- teragas L2
- amazing UX

- shorter slots
- precons
- faster finality

- FOCIL
- zkEVMs
- FCS

- \$1Q security
- optimality
- ossification



Tomasz's secret masterplan



	win	market
5		
4		
3		
2		
1	L2 confidence	Solana

Tomasz's secret masterplan



	win	market
5		
4		
3		
2	L2 network effects	Celestia
1	L2 confidence	Solana



	win	market
5		
4		
3	L1 unity	LayerZero
2	L2 network effects	Celestia
1	L2 confidence	Solana



	win	market
5		
4	store of value	BTC
3	L1 unity	LayerZero
2	L2 network effects	Celestia
1	L2 confidence	Solana



	win	market
5	global finance	???
4	store of value	BTC
3	L1 unity	LayerZero
2	L2 network effects	Celestia
1	L2 confidence	Solana

Tomasz's secret masterplan



	win	market	
5	global finance	???	} "lean"
4	store of value	BTC	
3	L1 unity	LayerZero	
2	L2 network effects	Celestia	} "mean"
1	L2 confidence	Solana	



social layer—believe in something™



social layer—believe in something™



	language dev count	coordinator location	website Github org
Zeam	Zig 2 devs	Gajinder India	zeam.in blockblaz
Ream	Rust 9 devs	Unnawut Thailand	reamlabs.com ReamLabs
[Quadrivium]	C++ 5 devs	Kamil Kazakhstan	qdrvm.io qdrvm
[Nethermind]	C# 2 devs	Jorge Spain	nethermind.io NethermindSil
[Kakarot]	Cairo 2 devs	Elias France	kakarot.org kkrt-labs
Lantern	C 5+ devs	Mihir Australia	piertwo.com Pier-Two
[LambdaClass]	Elixir or Rust 5 devs	Federico Argentina	lambdaclass.com lambdaclass
Colibri	C, C++ or Rust 2 devs	Jork Germany	corpus.io corpus-io
Afream	Rust 6 devs	Ayodeji Nigeria	afream.io Bluceducare

thank you :)

